In-distribution Public Data Synthesis with Diffusion Models for Differentially Private Image Classification

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Abstract

To alleviate the utility degradation of deep learning image classification with differential privacy (DP), employing extra public data or pre-trained models has been widely explored. Recently, the use of in-distribution public data has been investigated, where tiny subsets of datasets are released publicly. In this paper, we investigate a framework that leverages recent diffusion models to amplify the information of public data. Subsequently, we identify data diversity and generalization gap between public and private data as critical factors addressing the limited public data. While assuming 4% of training data as public, our method achieves 85.48% on CIFAR-10 with a privacy budget of $\varepsilon = 2$, without employing extra public data for training.

1. Introduction

Differential privacy (DP) [18, 19] establishes a mathematical framework to ensure the privacy of training data. In deep learning, differentially private SGD (DP-SGD) [1] has become the de facto standard method to guarantee the models' privacy. However, differentially private training inevitably degrades performance compared to standard (non-DP) training [6, 45]. As a practical solution, leveraging *public pre-trained models* or *public data* has been explored to enhance utility [12, 39, 59, 65–67]. As using public data raises no privacy concerns, fine-tuning pre-trained models on private data can make use of learned features for free. Nevertheless, their effectiveness might be diminished when addressing *out-of-distribution (OOD) public data* of small shared characteristics with private data [59].

As an alternative, researchers have investigated the use of *in-distribution (ID) public data*, indicating that a small portion of in-distribution data is made public [38]. For example, some data owners decide to share their data publicly in exchange for economic incentives or public interest. This setup allows us to leverage public data with a distribution similar to private data. In contrast to previous approaches, such as utilizing extra OOD data or distribution shift, we need to assume a limited amount of ID public data since the public data might contain more sensitive information. However, the repeated use of limited-sized public data can raise critical issues related to memorization and overfitting [43]. To mitigate the aforementioned problems, previous studies mainly have focused on utilizing side information from the little public data to enhance the optimization of differentially private deep learning [2, 4, 38, 43].

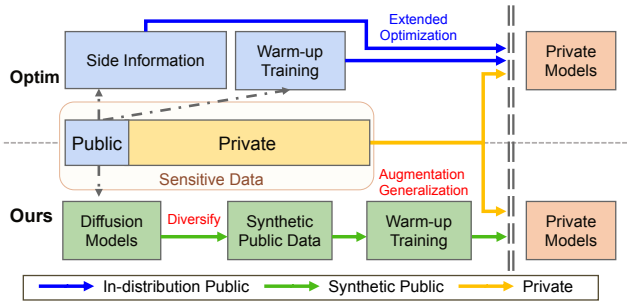


Figure 1. Illustration of (Top) existing optimization-based methods and (Bottom) proposed training procedure for amplifying in-distribution public data and utilizing the synthetic data.

Recently, Ganesh et al. [21] pointed out that warm-up training with public data plays a crucial in private optimization to find a good basin with a small loss, whereas DP-SGD from a random initial point faces a high loss and poor optimization. From this viewpoint, we suggest that enlarging the public dataset could improve the performance of private learning. By using the recent diffusion models [28, 30], we aim to enrich the contextual information of public data. We further identify that data diversity and generalization gap between public and private data act as critical factors, particularly addressing the limited size of public data.

The proposed framework is illustrated in Figure 1. [1] We summarize our approaches and their performance gains in

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[1]Our code is available at https://github.com/JinseongP/DPTrainer.

Table 1. Ablation study on the impact of various techniques, including synthesis, augmentation, and optimization, to enhance classification performance using in-distribution public data from CIFAR-10 under $(2, 10^{-5})$-DP. Refer to Section 3.1 for the details of setups and the relevant sections for the details of each method.

Setup	Training Settings	Test Acc
Baselines including previous SOTA methods (Sec. 3.1)		
Cold	Cold Baseline (WRN16-4) [12]	64.02%
Warm	Warm-up on public data	68.09%
WarmSyn	Warm-up on DDPM synthesis [43]	72.0%
WarmSE	WarmSyn (DDPM) + DOPE-SGD [43]	75.1%
In-distribution public data synthesis (Sec. 4.1) & Diversity (Sec. 4.2)		
WarmSyn	Warm-up on EDM synthesis	75.13 %
WarmSyn	EDM synthesis + generation diversity	77.66 %
WarmSyn	EDM synthesis + augmentation diversity	84.88%
Generalization for public-private datasets (Sec. 4.3)		
WarmSyn	Well-generalizing minimum during warm-up	85.48%

Table 1, assuming 4% of public data with a privacy budget of $\varepsilon = 2$ and $\delta = 10^{-5}$. The results outperform the existing state-of-the-art (SOTA) methods, i.e., boosting the accuracy from the previous 75.1% [43] to 85.48% on CIFAR-10. We emphasize that our approaches for dissecting warm-up training for DP are distinguishable from transfer tasks in non-DP setups, addressing data scarcity and improving underperforming DP-SGD optimization. Table 2 shows the gain and privacy cost of methods with public data usage compared to the training without public data, as in [21]. The results show higher gain and privacy costs with stronger privacy (lower ε), whereas ours shows the highest gain and smallest costs. Notably, we observe tiny improvements in non-private settings, highlighting the distinctiveness of DP scenarios and the unique strengths of our tailored approach.

Table 2. Gain (↑) in performance and the corresponding cost of privacy (↓, in parentheses) for each method compared to not exploiting public data. Refer to Section 3.1 for the details of the setups.

Methods	Non-priv.	$\varepsilon = 6$	$\varepsilon = 2$
Warm [43]	0.05% (-)	0.10% (18.80%)	3.20% (27.80%)
WarmSE [43]	0.13% (-)	3.00% (15.98%)	5.60% (25.48%)
Ours	0.13% (-)	**10.06% (8.92%)**	**20.58% (10.50%)**

2. Background and Related Work

2.1. Differentially Private Deep Learning

Differential privacy (DP) [19] can guarantee the privacy of training data as follows:

Definition 2.1 *(Differential privacy) For two adjacent inputs $d, d' \in \mathcal{D}$, a randomized mechanism $\mathcal{M} : \mathcal{D} \to \mathcal{R}$ satisfies (ε, δ)-differential privacy for any set of possible outputs $\mathcal{S} \subseteq \mathcal{R}$ if*

$$Pr[\mathcal{M}(d) \in \mathcal{S}] \leq e^{\varepsilon} Pr[\mathcal{M}(d') \in \mathcal{S}] + \delta. \quad (1)$$

The privacy budget $\varepsilon \geq 0$ controls the level of privacy guarantee with the broken probability $\delta \geq 0$. In deep learning, differentially private SGD (DP-SGD) [1] is widely used with the following steps: (i) average the clipped per-sample gradient $\nabla \ell_i(\boldsymbol{w}) \coloneqq \nabla \ell(\boldsymbol{w}; \boldsymbol{x}_i)$ for weight $\boldsymbol{w}$ with respect to each individual private data sample $\boldsymbol{x}_i \in \boldsymbol{X}_t^{pr}$ and (ii) add Gaussian noise to the averaged gradient as follows:

$$\begin{aligned} \boldsymbol{g}_t^{pr} &= \frac{1}{|\boldsymbol{X}_t^{pr}|} \sum_{\boldsymbol{x}_i \in \boldsymbol{X}_t^{pr}} \texttt{clip}\left(\nabla \ell_i(\boldsymbol{w}_t), C\right), \\ \tilde{\boldsymbol{g}}_t^{pr} &= \boldsymbol{g}_t^{pr} + \mathcal{N}(\boldsymbol{0}, C^2 \sigma^2 \mathbf{I}). \end{aligned} \quad (2)$$

The weight is updated as $\boldsymbol{w}_{t+1} = \boldsymbol{w}_t - \eta \tilde{\boldsymbol{g}}_t^{pr}$ with a learning rate η and $\texttt{clip}(\boldsymbol{u}, C)$ projects $\boldsymbol{u}$ to the L_2-ball of radius C. The noise level σ is determined by the privacy budget (ε, δ), the number of training steps, and the sampling probability (refer to Appendix C for details).

To guarantee privacy with clipping and noise addition, DP-SGD inevitably degrades the performance during optimization. Therefore, various studies explored DP-friendly properties, including architecture [9, 59], loss and activation functions [44, 54], or smoothness [46, 55, 60]. Notably, De et al. [12] introduced a new era in utilizing large models for DP-SGD tasks by introducing the averaged loss of various augmentations called augmentation multiplicity, weight standardization [49], and optimization tricks. Note that adaptive clipping [3, 8] could improve our results.

2.2. Private Learning with Public Information

The benefits of massive OOD public data have been explored in the context of the pre-trained models in vision [7, 12] and language [39, 65–67] models. Additionally, the transfer learning with similar distribution, such as CIFAR-100 for CIFAR-10, was under investigated [58, 59].

ID public data $\boldsymbol{X}^{pub}$ requires different approaches from the massive OOD public data due to its limited size and similar distributional properties. Hence, prior studies [2, 4, 38, 43], categorized as *extended methods* in Section 3.1, have focused on mitigating errors in the optimization of DP-SGD by utilizing the gradients of public data as a proxy for true gradients. These methods update the weight with $\boldsymbol{w}_{t+1} = \boldsymbol{w}_t - \eta f_g$, for $f_g(\boldsymbol{w}_t; \boldsymbol{X}_t^{pub}, \boldsymbol{X}_t^{pr})$ such as normalizing the private gradient with the public gradient norm [38], employing a linear combination of private and public gradients [2], or estimating low-rank approximation with public data (even unlabelled) to reduce noise levels [48, 65].

Nasr et al. [43] firstly introduced training a generative model as an augmentation technique. However, their primary focus is the optimization called DOPE-SGD, which updates towards the public gradient $\boldsymbol{g}_t^{pub}$ (without clipping and noise addition) and makes $\boldsymbol{g}_t^{pub} - \boldsymbol{g}_t^{pr}$ private as follows:

$$\boldsymbol{w}_{t+1} = \boldsymbol{w}_t - \eta(\boldsymbol{g}_t^{pub} + \texttt{clip}(\boldsymbol{g}_t^{pub} - \boldsymbol{g}_t^{pr}, C) + \mathcal{N}), \quad (3)$$

with the same noise level $\mathcal{N}$ in Equation (2). To the best of our knowledge, none of the prior studies deeply investigated the ID public data as a perspective of the data synthesis.

2.3. Diffusion Synthesis

The integration of generative models in classification tasks has been widely explored to enhance generalization performance without extra data samples [5, 22, 25]. In recent years, diffusion models [28, 56, 57] have shown promising results in image generation. Notably, Xiao et al. [62] argued that diffusion models can generate high-quality and diverse images. Subsequently, various studies have focused on enhancing the quality of generation [30, 32], conditioning sampling [15, 27], and text-to-image generation [50, 52].

Due to the significance of privacy concerns during image synthesis, differentially private diffusion synthesis is one of the actively investigated research topics [17, 41]. Given the effectiveness of private fine-tuning upon pre-trained diffusion models for generation and classification tasks [23], utilizing privacy-preserving diffusion models presents opportunities for our future works.

Measures To measure the quality of generated images, various measures are proposed: Inception Score (IS) [53] and Precision [36] to measure fidelity, Recall [36] to measure diversity, and Fréchet Inception Distance (FID) [26] to measure the distributional quality for mean and variance.

To assess the impact of synthetic data on classification tasks, Ravuri and Vinyals [51] proposed the Classification Accuracy Score (CAS), which evaluates the classification performance on a test set using a model trained on synthetic data. Thus, we use CAS to validate the performance of synthetic public data on test data. However, since our main focus is on improving classification performance including private learning, all of the previous measures cannot be perfectly aligned with our goals.

3. Framework

3.1. Training Procedure

Our objective is to investigate the potential of diffusion models with ID public data, departing from conventional approaches that focus on guiding optimization with the public data [2, 4, 38, 43, 48, 64]. Our work aligns closely with [43], but we specifically concentrate on synthesizing data from the public and examining the important properties of private learning. We summarize the training scenarios using ID public data as follows [43]:

- **Cold**: Conduct DP training (e.g., DP-SGD) on the private dataset without using public data.
- **Warm**: Train models through (i) non-DP *warm-up training* phase on the public dataset (e.g., SGD) and (ii) *private training* phase on the private dataset with DP methods.

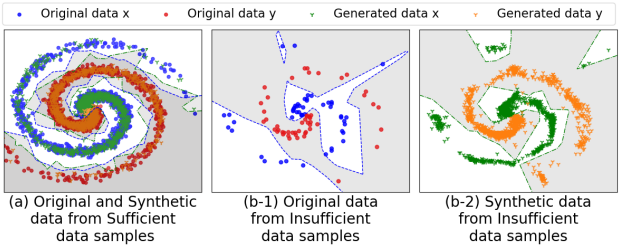


Figure 2. Toy experiment on diffusion synthesis and its effect on decision boundaries while varying the training sample size.

- **WarmExt**: During the private training (after warm-up training), leverage the public side information, such as gradients [2, 4, 38, 43], or low-rank approximation [48, 64], as *extended methods* of DP optimization.
- **WarmSyn** (**Ours**, [43]): Amplify the sparse information of public data using generative models or augmentation methods. Optionally, **WarmSE** indicates employing extended optimization with synthetic public data.

In terms of privacy, since utilizing public data does not pose any privacy concerns to the private data, we can leverage the public data multiple times, e.g., training warm-up classification models, diffusion models, or guiding private training. From now on, we denote in-distribution (ID) public data as public data unless otherwise specified.

3.2. Public Data Synthesis with Diffusion Models

For amplifying the sparse information in the WarmSyn setting, we first investigate amplifying the number of data samples using diffusion models. Recent studies [5, 22, 25, 61] have demonstrated the performance gain in complicated classification problems by training diffusion models to expand the size of training data samples within the data manifold, e.g., amplifying 50K CIFAR-10 training data into a 50M synthetic dataset improves adversarial training [61].

To analyze public data synthesis with limited data samples, we conduct an experiment using a spiral dataset as a toy example (detailed in Appendix C), as shown in Figure 2. We consider two settings: (a) with a sufficient and (b) with an insufficient number of original data samples. For each setting, we train a simple diffusion model to generate new data samples. Then, we train two-layer classifiers until convergence using the original and synthetic data, respectively.

With a large number of samples in (a), both diffusion and classification models demonstrate well-trained results. However, when the original data size is limited as in (b-1), the classification model tends to overfit individual data. Thus, despite correctly classifying almost all data samples, the model fails to discover the proper decision boundary. In contrast, in (b-2) where the diffusion model effectively approximates the data manifold, the classifier achieves a superior decision boundary compared to (b-1). Nevertheless, we should be aware that diffusion models can be apt to over-

Table 3. Quality comparison of synthetic data trained with 4% of public data on CIFAR-10.

Sampling	Fidelity		Diversity	Quality	CAS (↑)	Test Acc
	IS (↑)	Precision (↑)	Recall (↑)	FID (↓)	(%)	(%)
EDM (w_d=0)	**11.008**	**0.964**	0.157	7.799	62.82	75.13
EDM + DG (w_d=3)	10.815	**0.964**	0.153	**7.786**	62.47	75.31
EDM + DG (w_d=10)	10.796	0.946	0.170	8.274	64.61	75.98
EDM + DG (w_d=20)	10.157	0.873	0.191	11.497	**67.61**	**77.66**
EDM + DG (w_d=30)	9.113	0.785	**0.211**	19.748	66.53	77.22



Figure 3. Selected synthetic data samples from (Top) EDM and (Bottom) EDM + DG (w_d = 30) synthesis.

and underestimate the data distribution, as shown in (b-2).

To push further, we mathematically analyze how the rate of data samples affects synthesis. The ability of a generative model θ to approximate the 'entire' data can be determined by the ratio of 'seen' (observed during training) and the 'unseen' (not observed during training). Specifically, with a limited number of seen data, the model is required to estimate the unseen data to approximate the entire data.

Theorem 3.1 *For a finite number of 'entire' data samples $S_{data} = \{\boldsymbol{x}_1, \cdots, \boldsymbol{x}_N\}$, split the data samples into 'seen' data $S_s = \{\boldsymbol{x}_1, \cdots, \boldsymbol{x}_n\}$, and 'unseen' data $S_u = \{\boldsymbol{x}_{n+1}, \cdots, \boldsymbol{x}_N\}$, without loss of generality. Let p_{data}, p_s, and p_u be the probability distribution of each corresponding dataset. For any generative model θ trained on 'seen' data, p_θ represents the data distribution generated by θ. Then,*

$$\log \frac{1}{r_1} + \hat{p}_\theta^u \log \frac{\hat{p}_\theta^u}{r_2} \leq \tilde{D}_{KL}(p_\theta \| p_{data}) - \tilde{D}_{KL}(p_\theta \| p_s) \leq \log \frac{1}{r_1} + \hat{p}_\theta^u \log \frac{1}{r_2},$$

where $\hat{p}_\theta^u = \sum_{\boldsymbol{x} \in S_u} p_\theta(\boldsymbol{x}) \leq 1$ indicates the capacity of the model θ to generate 'unseen' data. For discrete probability distributions p and q, $\tilde{D}_{KL}(p\|q)$ indicates the relaxed Kullback–Leibler (KL) divergence, excluding the case when $q = 0$. $r_1 = \frac{n}{N}$ and $r_2 = \frac{N-n}{n}$ denote the ratios of 'seen' to 'entire', and 'unseen' to 'seen' data, respectively.

The detailed proof is presented in Appendix A. The above theorem indicates that the ratios r_1 and r_2 affect how the model θ can approximate the p_{data}. As $r_1 \to 1$ with sufficient seen data, then $r_2 \to 0$ and $S_u \to \varnothing$, so that $\hat{p}_\theta^u \to 0$ and the upper and lower bounds converge to zero. Thus, ensuring p_θ is close to p_s ensures proximity of p_θ to p_{data}.

However, when considering a small r_1, which is equivalent to a limited number of public data setups, the bounds depend on $\hat{p}_\theta^u$. For $r_1 \leq \frac{1}{e+1}$, the lower and upper bounds are decreasing functions with respect to $\hat{p}_\theta^u$. This implies that as the model's capacity to generate unseen data increases, the differences in divergences decrease. Thus, approximating the distribution of seen data is not enough to learn the whole data distribution for small r_1. Rather, it is important for the model to learn the data manifold effectively to better approximate the data distribution including the unseen data.

4. Generated Data for Private Learning

We now investigate how the aforementioned diffusion synthesis can be utilized in the context of public data and analyze how it can be further leveraged to enhance performance. For experiments, we primarily focus on the CIFAR-10 dataset with a privacy budget of $(2, 10^{-5})$, utilizing WRN-16-4[2] with 16 augmentation multiplicity and the techniques proposed in [12]. Within the training set, we randomly select 2K instances (4%) for the public samples, as suggested in [43]. These samples are uniformly drawn from each class. All the measures for evaluating the generated data in this section are calculated with the entire training set (50K) as a reference. Additional experimental details are provided in Section 5.1 and Appendix B.

4.1. Better Diffusion Synthesis with Public Data

Theorem 3.1 demonstrates that well-approximating the distribution of seen public data p_s enables the synthetic data distribution p_θ mimic the entire data distribution p_{data}, which is essential to obtain good decision boundary as in Figure 2. Thus, we initially replace the Denoising Diffusion Probabilistic Model (DDPM) [28], previously investigated by Nasr et al. [43] for public synthesis, with the Elucidating Diffusion Model (EDM) [30]. EDM is known to generate better images (e.g., lower FID scores) than DDPM by using a higher-order sampling process. Then, we train EDM on 2K public data samples without using external datasets and employ class-conditional sampling to match the original distribution. With the trained EDM, we calculate various measures in Table 3. The FID of EDM synthesis at 7.80 (correspondingly 7.89 on 40K) outperforms the reported FID of 12.8 of DDPM on 40K images [43], which also improves the classification performance as shown in Table 1.

However, the generation quality using public data is notably worse than the FID of 1.79 achieved with the entire set [30]. Within a limited public data, the model struggles to capture diversity, resulting in a recall of 0.16, even though precision remains high at 0.96. Note that repeating each public sample 25 times (thus 50K samples) results in a precision of 1.00, recall of 0.04, and FID of 13.64. The generated images and their memorization are illustrated in Ap-

[2]WRN-40-4 requires excessive GPU considering its performance [43].

Table 4. CAS (%), test accuracy (%), and their difference (%) with different augmentation methods during warm-up phase.

Augmentation	CAS(%)	Test Acc (%)	Diff (%)
No Aug (EDM)	62.82	75.13	13.31
Common	77.99	83.97	5.98
Common + Cutout	**80.72**	**84.88**	4.16
Common + Cutmix	65.73	82.20	**16.47**
AutoAugment	77.21	83.45	6.24

Table 5. Geometric measures of the models trained with SGD and SAM after warm-up phase.

Opt	Synthetic Data			Private Data		
	λ_{max}	λ_{max}/λ_5	$\mathrm{Tr}(\nabla^2\ell(\boldsymbol{w}))$	λ_{max}	λ_{max}/λ_5	$\mathrm{Tr}(\nabla^2\ell(\boldsymbol{w}))$
SGD	1.38	10.62	71.98	112.82	1.58	-2527.78
SAM	**0.44**	**2.63**	**53.21**	**58.32**	**1.50**	333.03

pendix E. No privacy concern occurs in diffusion synthesis since the diffusion model is solely trained with public data.

4.2. Data Diversity Matters for Public Data

With limited private data, the model's capacity to generate unseen data is required in Theorem 3.1. The lack of diversity in the generated data can lead to misclassification due to distorted decision boundaries, as described in Figure 2. Given the challenge of DP-SGD in identifying a good basin [21], the key to successful warm-up lies in finding a well-generalizing minimum with a high CAS. As observed in [25, 51], data diversity can play an important role in improving CAS. We hypothesize that the importance of diversity is more pronounced in the warm-up phase of private learning, where the generated images exhibit low recall values.

Diversity for generation To validate our hypothesis, we first investigate enforcing data diversity in the generation process. Obtaining diversity in generations solely through a diffusion model is a challenging task. Thus, to enhance the data diversity during generation, we can use variants of diffusion models for guidance [27, 32] or editing [10, 40, 42]. Among them, discriminator guidance (DG) [32] introduces a discriminator to judge whether the sampling is from the true data or synthesis, controlling the trade-off between fidelity and diversity of the generated images by adjusting the weight w_d of the discriminator. Higher w_d yields diversity. Refer to Appendix C for the details of DG.

As shown in Table 3, a higher weight of guidance ensures greater data diversity without any augmentation, but sacristies the fidelity. The best FID score is obtained with $w_d = 3$ while the best CAS is obtained at a bigger weight $w_d = 20$. Interestingly, we need a larger weight value to achieve a similar gain of diversity than standard training ($w_d = 1.5$ for best FID [32]). Figure 3 represents selected examples from two extreme cases, with $w_d = 0$ and $w_d = 30$ to visualize the difference. Despite the quality degradation of detailed features with $w_d = 30$, the CAS is higher due to the increased data diversity in features. Thus, we need to enhance diversity while maintaining quality for better classification. Similar to the EDM model trained with public data, no privacy concerns occur since the discriminator requires only original public and synthetic data from EDM.

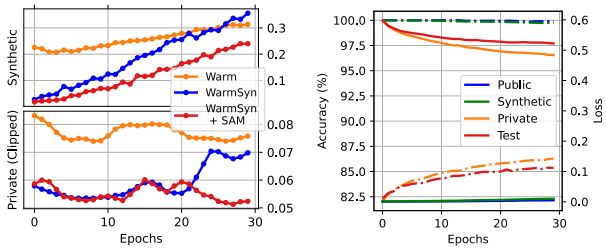
Diversity with augmentation Nonetheless, training an additional discriminator for DG can pose a bottleneck in terms of time and computational efficiency. Thus, to explicitly enhance the diversity of EDM synthetic data during the warm-up training, we employ various data augmentation techniques designed for classification tasks. Wang et al. [61] argued that utilizing appropriate augmentation in diffusion-based generated images can further improve the classification performance. Common augmentation [24] uses padding and random crop to the original size and horizontal flipping for the images. Cutmix [68] randomly replaces a part of the image with another and Cutout [14] randomly pad images. AutoAugment [11] chooses the best combination of augmentations such as color, rotation, or cutout. We set a baseline with no augmentation for synthetic images of EDM and compare these augmentations.

The results are summarized in Table 4. Adding cutout augmentation to common augmentation demonstrates the best performance in terms of CAS and test accuracy. However, the two measures are not always aligned. Rather, the best performance gain with private data, noted as *Diff (%) = Test Acc−CAS*, is obtained from Cutmix, which is known as one of the most diversified augmentations. This indicates that robust decision boundaries achieved through diversity facilitate easier private learning. Interestingly, the performance gain of augmentation surpasses the benefits of generation diversity, as illustrated in Appendix C.

4.3. Optimization with Synthetic Data

As the clipping of DP-SGD significantly lowers the performance, it is known that a smaller gradient norm ensures high performance in private training [44, 46, 54]. In Figure 4a, we illustrate the gradient norms during private training with different training setups. The norm of synthetic data is calculated without clipping, whereas the private data used for DP-SGD is summed over individually clipped gradients, each with a clipping value of $C = 1$. The results indicate that utilizing a synthetic dataset reduces the gradient norm, even in a private dataset where constraining the gradient norm is crucial to mitigate the effect of clipping.

However, even though training with diversified synthetic public data and augmentation, the models face the risk of being overfitted to public data. As public samples are selected from in-distribution data, the problem of the public-



(a) Gradient norm. (b) Accuracy (−·) and loss (−).

Figure 4. Learning dynamics of private training with different datasets and optimization methods.

private generalization gap is analogous to the training-test generalization gap in standard training. In Figure 4b, we illustrate the accuracy and loss for each public, synthetic, private, and test dataset during private training after the warm-up phase. The model consistently achieves near-zero loss and 100% accuracy on synthetic data (as well as on public data) but struggles with private data (as well as on test data).

To alleviate the generalization gap in standard training, various studies [16, 31, 33, 34, 37, 47] explored the geometric properties of the loss function in the weight space. The prominent optimization methods are designed to uncover flat minima, such as sharpness-aware minimization (SAM) [20] or stochastic weight averaging (SWA) [29]. Therefore, we apply SAM during the warm-up training and calculate geometric generalization measures, including metrics such as the top Hessian value λ_{max}, the ratio of Hessian λ_{max}/λ_5, and the trace of the Hessian matrix $\mathrm{Tr}(\nabla^2\ell(\boldsymbol{w}))$, for both synthetic and private data. Table 5 indicates that utilizing SAM results in lower (better) metric values, not only for synthetic data but also for private data. Note that SGD even demonstrates a negative trace value on private data.

Moreover, in Figure 4a, as the flat minimum of SAM leads to a decreased gradient norm on synthetic data, it also maintains similar results concerning private data. These results indicate that relieving overfitting to synthetic data can help the generalization of private data. Thus, the aforementioned approaches find suitable initial point of DP-SGD.

5. Experiments

5.1. Experimental setup

We assess the effectiveness of our proposed methods using public data primarily in two datasets: CIFAR-10 and CIFAR-100. For the public dataset, we randomly sample 4% of the training data (2K samples) uniformly drawn from each class, while the remaining data are used as private samples following [43]. We then train the EDM [30] models with the 2K public data samples and build 50K synthetic datasets with EDM sampling. To address the computational burden of training diffusion and sampling, we generate datasets once and reuse them for classifiers as in [43].

For classification models, we adopt WRN-16-4, following the techniques in [12] with 16 augmentations, and use pre-trained vision transformer models following [7]. Our experiments are conducted using PyTorch libraries [63] on eight NVIDIA GeForce RTX 3090 GPUs, partially on a cloud server with four NVIDIA A100 40GB GPUs. We report the mean and standard deviation for each experiment. For the detailed settings, refer to Appendix B.

5.2. Classification Performance with Public Data

Effects of individual techniques We first revisit Table 1, the ablation study of sequentially employing our approaches, with a privacy budget $(2, 10^{-5})$-DP on CIFAR-10. By only using better EDM synthesis without extra optimization techniques, we achieve the previous SOTA performance of 75.1% [43]. Additionally, recognizing the significance of diversity in data generation, we employ cutout augmentation techniques for diversity in classification, resulting in a performance of 84.88%. To mitigate potential overfitting to public information, we make use of generalization techniques. All these efforts collectively lead to an accuracy of 85.48% under $(2, 10^{-5})$-DP.

Despite achieving the best performance of 85.93% when combined with DG, we exclude DG in this section due to its marginal performance enhancement (0.45%p) considering its extended training time, as detailed in Appendix C.

CIFAR-10 with public data We report the performance comparison for our method with various previous approaches on a wide range of $\varepsilon \in \{1, 2, 3, 4, 6\}$ with $\delta = 10^{-5}$ in Table 6. Our approach, which incorporates EDM synthesis, augmentation, and optimization for the public data, exhibits superior classification performance when compared to existing methods including DDPM-based augmentation and extended optimizations. The accuracies of $\varepsilon = 0$ (after warm-up, without private data) and $\varepsilon = \infty$ (not private, with clipping) are 80.72% and 88.52%, respectively.

CIFAR-100 with public data We then explore the CIFAR-100 dataset for DP, which is not actively investigated without pre-trained models due to its complexity. We first train the EDM model with 2K images and generate 50K images, similar to CIFAR-10. Given the 100 classes in CIFAR-100, only 20 public samples are available per class, significantly fewer than in CIFAR-10. The FID on 50K images with EDM synthesis is 11.28, which reduces the original FID of 15.58 for replicating each public image 25 times. After the warm-up, the accuracy of synthetic images on $\varepsilon = 0$ is 38.04% (46.74% on $\varepsilon = \infty$), while the test accuracy of using 2K public samples without synthesis on $\varepsilon = 0$ is only 16.79%. In Table 7, we report the accuracies by sequentially adopting the aforementioned tech-

Table 6. Test accuracy (%) of private training on CIFAR-10 with privacy budgets of $\varepsilon \in \{1, 2, 3, 4, 6\}$. The public and synthesis columns denote the Warm and WarmSyn setups, respectively. Ours employs all the techniques in Table 1, i.e., synthesis, augmentation, and optimization. We highlight the best accuracy in **bold**.

Datasets	Architecture	Public	Synthesis	Method	$\varepsilon = 1$	$\varepsilon = 2$	$\varepsilon = 3$	$\varepsilon = 4$	$\varepsilon = 6$
CIFAR-10	CNN-Tanh (0.55M)	✗	✗	[44]	45.8	58.3	63.5	-	-
	ScatterNet (0.16M)	✗	✗	[59]	60.3	67.2	69.3	-	-
	DPNAS [9] (0.53M)	✗	✗	[46]	60.13±0.34	67.23±0.12	69.86±0.49	-	-
	WRN-16-4 (2.74M)	✗	✗	[12]	56.8±0.6	64.9±0.5	69.2±0.3	71.9±0.3	77.0±0.8
	WRN-40-4 (8.94M)	✗	✗	[12]	56.4±0.6	65.9±0.5	70.7±0.2	73.5±0.6	78.8±0.4
	WRN-16-4 (2.74M)	✓	✗	[2]†	-	68.7	-	73.1	77.2
		✓	✗	[38]†	-	68.7	-	73.5	77.9
		✓	✓	[2]†	-	70.5	-	74.5	78.2
		✓	✓	[38]†	-	69.1	-	74.1	78.1
		✓	✓	[43]†	-	75.1	-	77.9	80.0
		✓	✓	**Ours**	**84.30±0.11**	**85.48±0.12**	**86.03±0.09**	**86.49±0.13**	**87.06±0.24**

†We note the results reported in [43] to set the architecture same. All other baseline results are adopted from the original paper.

Table 7. Test accuracy (%) of private classification on CIFAR-100 on the privacy budget of $\varepsilon \in \{1, 2, 6, 10\}$. The public and synthesis columns indicate Warm and WarmSyn settings, respectively. We employ the techniques in Table 1 sequentially, i.e., synthesis, augmentation, and optimization. We highlight the best accuracy in **bold**.

Datasets	Architecture	Public	Synthesis	Methods	$\varepsilon = 1$	$\varepsilon = 2$	$\varepsilon = 6$	$\varepsilon = 10$
CIFAR-100	Resnet-9	✗	✗	[35]†	18.1	24.9	-	40.8
	Resnet-9 (6.62M)	✗	✗	Cold†	8.35	14.42	29.89	35.11
	WRN-16-4 (2.74M)	✗	✗	Cold	9.28	18.19	33.61	39.09
	WRN-16-4 (2.74M)	✓	✗	Warm	20.84	25.15	33.47	38.89
		✓	✓	WarmSyn	26.13±0.20	31.53±0.04	35.53±0.11	40.82±0.10
		✓	✓	+Augmentation	40.96±0.30	44.52±0.01	50.47±0.04	54.29±0.05
		✓	✓	+Optimization	**45.93±0.06**	**48.61±0.59**	**54.36±0.52**	**56.56±0.11**

†Unfortunately, their DP-SGD results are not reproducible, even when using the same hyperparameters as in the original paper.

niques, where each of the approaches matters. As a result, we obtain 48.61% on $\varepsilon = 2$ without pre-trained models.

CIFAR-100 with pre-trained on ImageNet To push further, we demonstrate the effectiveness of our procedures when combined with pre-trained models with ID public data. We adopt the vision transformers following [7] pre-trained on ImageNet [13]. Upon the pre-trained models, we sequentially perform the warm-up training with ID public data and private learning. The results in Table 8 indicate that the synthesis can boost classification performance. Within a wide range of models and privacy budgets, our methods outperform the warm settings.

Distribution shift (CIFAR-100 → CIFAR-10) The distribution shifts from similar but not ID public data [59] can be combined with our methods. We consider 4% of CIFAR-100 data as public to enhance the performance of CIFAR-10 as a substitute for CIFAR-10 ID public data. In Table 9, we present the classification results for both the warm and our settings, where both models are trained on the public data and synthetic data from 4% of CIFAR-100, respectively. The experimental results show that the differences in accuracies exceed 20%, even though neither model has observed CIFAR-10 datasets before private training with DP-SGD. This indicates that amplifying public information is meaningful in distribution shifts. Additional sensitivity analysis and ablation studies can be found in Appendix D.

Pre-trained diffusion models Our main idea to use synthesis for public data is flexible in practical scenarios and diverse datasets. To mitigate the drawbacks of training diffusion on each dataset, we investigated using the pre-trained diffusion models, which are trained on OOD public data. Specifically, we make use of pre-trained Stable Diffusion (SD) [52] and Boomerang sampling [40]. We fed 4% of public data to Boomerang editing with SD backbone for generation diversity and generated 20 images per sample for classifier training. The similar experiments are shown in Appendix D.We maintain the other training procedures the same as 'Ours', which we call 'Ours-SD'. Table 10 indicates that our framework is straightforward and effective for real-world data, and generalizable without depending on specific diffusion models. We conclude that data diversity can enhance performance when the model is not trained on specific ID public data.

Table 8. Test accuracy (%) of private classification on CIFAR-100 using pre-trained models on the privacy budget of $\varepsilon \in \{0.5, 2\}$. The **bold** indicates results within the standard deviation of the top mean score.

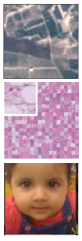
Datasets	Privacy budget	$\varepsilon = 0.5$			$\varepsilon = 2$		
	Architecture	Cold	Warm	Ours	Cold	Warm	Ours
CIFAR -100	CrossViT small 240 (26.3M)	60.50±1.70	73.17±0.24	**77.43±0.13**	71.36±0.52	77.00±0.27	**80.25±0.57**
	CrossViT 18 240 (42.6M)	67.03±0.46	77.19±0.44	**79.96±0.10**	74.28±0.22	80.30±0.02	**82.85±0.08**
	DeiT base patch16 224 (85.8M)	49.34±1.49	**79.56±0.74**	**79.28±0.76**	69.09±0.10	82.98±0.04	**83.06±0.01**
	CrossViT base 240 (103.9M)	66.24±0.47	75.59±0.21	**78.06±0.05**	75.15±0.03	79.25±0.05	**81.14±0.15**

Table 9. Comparison of distribution shift performance on CIFAR-10 with the warm and our models trained on 4% of CIFAR-100.

CIFAR-100 → CIFAR-10	Test Acc (%)		
	$\varepsilon = 2$	$\varepsilon = 4$	$\varepsilon = 6$
Warm	50.65±0.37	56.54±0.20	59.49±1.39
Ours	**73.46±0.28**	**78.60±0.13**	**80.12±0.04**

Table 10. Classification results on real-world datasets.

Data	Method	$\varepsilon = 2$	$\varepsilon = 4$	$\varepsilon = 6$
EuroSAT (Land use)	Warm (w/ Aug, Opt)	83.67	85.52	86.07
	WarmExt [43]	84.37	85.78	86.07
	Ours-SD	**90.37**	**91.74**	**92.52**
PathMNIST (Biomedical)	Warm (w/ Aug, Opt)	89.23	89.39	89.54
	WarmExt [43]	89.04	89.25	89.28
	Ours-SD	**91.64**	**91.94**	**92.10**
FairFace (w.r.t. Race)	Warm (w/ Aug, Opt)	53.00	54.55	54.83
	WarmExt [43]	53.33	54.74	54.84
	Ours-SD	**54.90**	**56.19**	**56.37**



5.3. Revisit Extended Optimization

On top of our method, we reevaluate the effectiveness of existing extended methods (thus WarmSE) in Table 11, particularly mirror GD [2] and DOPE-SGD [43]. These methods are trained after our synthesis, augmentation, and warm-up optimization on $\varepsilon \in \{2, 4, 6\}$. Remarkably, the performance of the extended methods falls within the standard deviation range of DP-SGD. This implies that, as the models have already extracted substantial side information from public data before private training, extended optimization does not further improve DP optimization. Note that the computational time of the public batch is marginal to DP-SGD.

However, extended methods sometimes exhibit poorer training stability than DP-SGD. Due to their reliance on public gradients, these methods are sensitive to the hyperparameter settings and easily encounter exploding gradients. In Figure 5, under the same setting as Figure 4a, we observe a gradual decrease in the private gradient norm, while the public gradient norm consistently increases. When the public norm increases to a certain level, the norm of both public and private data with extended methods may diverge after updating toward public gradients. For further insights about the learning dynamics in terms of the loss function and private gradient norm without clipping, please refer to Appendix C. However, we believe that undiscovered extended methods can be beneficial for the WarmSyn settings.

Table 11. Performance and computational time of different extended optimization methods.

Privacy budget ε ($\delta = 10^{-5}$)	Optimization				
	DP-SGD	Mirror GD [2]		DOPE-SGD [43]	
		Median	Min	Median	Min
$\varepsilon = 2$	85.48±0.17	85.52	10.00	85.53	10.00
$\varepsilon = 4$	86.49±0.13	86.31	10.00	86.69	10.00
$\varepsilon = 6$	87.06±0.24	86.73	10.00	86.84	10.00
Time (ms/image)	12.70	12.80		12.86	

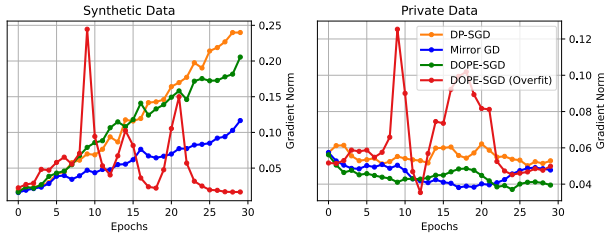


Figure 5. The norm of gradients with (Left) synthetic dataset and (Right) private dataset with clipping during private training with different optimization methods.

6. Conclusion

In this paper, we investigate the potential of diffusion models for in-distribution public data to enhance private classification performance. We demonstrate the importance of synthetic data diversity, augmentation techniques, and the importance of well-generalizing minima for private optimization. As a limitation, we leave the experiments on more extensive and sensitive datasets and the efficient implementation of our method. We hope that this work helps leverage diffusion models to address utility-privacy trade-offs.

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In-distribution Public Data Synthesis with Diffusion Models for Differentially Private Image Classification

Supplementary Material

A. Proofs

A.1. Proof of Theorem 3.1.

Following [12], we address this theorem with a finite number of samples $S_{data} = \{\boldsymbol{x}_1, \cdots, \boldsymbol{x}_N\}$. As mentioned in Appendix B.3 of [12], p_{data} of 'entire' data and p_s of 'seen' data can be represented by mixtures of Dirac delta distributions:

$$p_{data}(\boldsymbol{x}) = \frac{1}{N}\sum_{i=1}^{N}\delta(\boldsymbol{x}-\boldsymbol{x}_i),$$

and

$$p_s(\boldsymbol{x}) = \frac{1}{n}\sum_{i=1}^{n}\delta(\boldsymbol{x}-\boldsymbol{x}_i),$$

where $\boldsymbol{x} \in \mathcal{X}$ which is discrete sample space, since any pixel of an image can be represented by int between 0 to 255. p_θ can be defined in $\mathcal{X}$, which contains S_{data}. We defined the measure based on the KL divergence without $q(\boldsymbol{x}) = 0$ for the discrete probability, similar to [19], as follows:

$$\tilde{D}_{KL}(p\|q) = \sum_{\substack{\boldsymbol{x}\in\mathcal{X},\\ q(\boldsymbol{x})\neq 0}} p(\boldsymbol{x})\log\frac{p(\boldsymbol{x})}{q(\boldsymbol{x})}.$$

$\tilde{D}_{KL}$ becomes the KL divergence when the support of p is included in the support of q [9]. Then, by the definition:

$$\begin{aligned}
\tilde{D}_{KL}(p_\theta\|p_{data}) &= \sum_{\boldsymbol{x}\in S_{data}}\left(p_\theta(\boldsymbol{x})\log\frac{p_\theta(\boldsymbol{x})}{p_{data}(\boldsymbol{x})}\right)\\
&= \sum_{\boldsymbol{x}\in S_s}\left(p_\theta(\boldsymbol{x})\left[\log\frac{p_\theta(\boldsymbol{x})}{p_s(\boldsymbol{x})}+\log\frac{p_s(\boldsymbol{x})}{p_{data}(\boldsymbol{x})}\right]\right)\\
&\quad+\sum_{\boldsymbol{x}\in S_u}\left(p_\theta(\boldsymbol{x})\log\frac{p_\theta(\boldsymbol{x})}{p_{data}(\boldsymbol{x})}\right)\\
&= \tilde{D}_{KL}(p_\theta\|p_s)+\log\frac{N}{n}\sum_{\boldsymbol{x}\in S_s}p_\theta(\boldsymbol{x})\\
&\quad+\sum_{\boldsymbol{x}\in S_u}\left(p_\theta(\boldsymbol{x})\log\frac{p_\theta(\boldsymbol{x})}{p_{data}(\boldsymbol{x})}\right)\\
&= \tilde{D}_{KL}(p_\theta\|p_s)+\log\frac{N}{n}(1-\sum_{\boldsymbol{x}\in S_u}p_\theta(\boldsymbol{x}))\\
&\quad+\sum_{\boldsymbol{x}\in S_u}p_\theta(\boldsymbol{x})\log(Np_\theta(\boldsymbol{x}))\\
&= \tilde{D}_{KL}(p_\theta\|p_s)+\log\frac{N}{n}+\log n\sum_{\boldsymbol{x}\in S_u}p_\theta(\boldsymbol{x})\\
&\quad+\sum_{\boldsymbol{x}\in S_u}p_\theta(\boldsymbol{x})\log p_\theta(\boldsymbol{x}).
\end{aligned}$$

Note that as $N \to |\mathcal{X}|$, i.e., $S_{data} \to \mathcal{X}$, the support of p_{data} contains the support of p_θ so that $\tilde{D}_{KL}$ becomes the same as the (reverse) KL divergence. Since $x\log(x)$ is convex, we can use Jensen's inequality. For convenience, let $\bar{p}^u_\theta := \frac{1}{N-n}\sum_{\boldsymbol{x}\in S_u}p_\theta(\boldsymbol{x}) = \frac{1}{N-n}\hat{p}^u_\theta \le \frac{1}{N-n}$. Then,

$$\begin{aligned}
&\tilde{D}_{KL}(p_\theta\|p_{data})-\tilde{D}_{KL}(p_\theta\|p_s)\\
&\quad= \log\frac{N}{n}+\log n\sum_{\boldsymbol{x}\in S_u}p_\theta(\boldsymbol{x})+\sum_{\boldsymbol{x}\in S_u}p_\theta(\boldsymbol{x})\log p_\theta(\boldsymbol{x})\\
&\quad= \log\frac{N}{n}+(N-n)\bar{p}^u_\theta\log n+\sum_{\boldsymbol{x}\in S_u}p_\theta(\boldsymbol{x})\log p_\theta(\boldsymbol{x})\\
&\quad\ge \log\frac{N}{n}+(N-n)\bar{p}^u_\theta\log n+(N-n)\bar{p}^u_\theta\log\bar{p}^u_\theta\\
&\quad= \log\frac{N}{n}+(N-n)\bar{p}^u_\theta\log n\bar{p}^u_\theta\\
&\quad= \log\frac{N}{n}+\hat{p}^u_\theta\log\frac{n}{N-n}\hat{p}^u_\theta.
\end{aligned}$$

For small $n \le \frac{N}{e+1}$, $\bar{p}^u_\theta\log n\bar{p}^u_\theta$ is a decreasing function in $0 \le \bar{p}^u_\theta \le \frac{1}{N-n}$. Therefore, the lower bound is a decreasing function with respect to $\bar{p}^u_\theta$. Meanwhile,

$$\begin{aligned}
&\log\frac{N}{n}+\log n\sum_{\boldsymbol{x}\in S_u}p_\theta(\boldsymbol{x})+\sum_{\boldsymbol{x}\in S_u}p_\theta(\boldsymbol{x})\log p_\theta(\boldsymbol{x})\\
&\quad= \log\frac{N}{n}+\sum_{\boldsymbol{x}\in S_u}p_\theta(\boldsymbol{x})\log np_\theta(\boldsymbol{x})\\
&\quad\le \log\frac{N}{n}+\sum_{\boldsymbol{x}\in S_u}p_\theta(\boldsymbol{x})\log\frac{n}{N-n}\\
&\quad= \log\frac{N}{n}+(N-n)\bar{p}^u_\theta\log\frac{n}{N-n}\\
&\quad= \log\frac{N}{n}+\hat{p}^u_\theta\log\frac{n}{N-n}.
\end{aligned}$$

For small $n < \frac{N}{2}$, $\log\frac{n}{N-n} < 0$, and the upper bound is a decreasing function with respect to $\bar{p}^u_\theta$. Note that as $n \to N$, $p_{pub} \to p_{data}$, and the upper bound and the lower bound achieve equality.

B. Experimental settings

For experiments, we follow the learning framework outlined in Algorithm 1.

Algorithm 1: Private training with public data

Input: Private dataset $\boldsymbol{D}^{pr}$, public dataset $\boldsymbol{D}^{pub}$; Initial classifier $\boldsymbol{w}$, diffusion model $\boldsymbol{\theta}$, (discriminator ψ); Synthetic dataset size k.
Output: Final private classifier $\boldsymbol{w}_T$.
(Phase I) Synthesis and warm-up training
Train diffusion with public $\boldsymbol{\theta}^* \leftarrow (\boldsymbol{\theta}; \boldsymbol{D}^{pub})$.
if *Generation diversity* **then**
  Train discriminator $\psi^* \leftarrow (\psi; \boldsymbol{\theta}^*, \boldsymbol{D}^{pub})$.
Make synthetic public set $\boldsymbol{D}^{syn} \leftarrow (\boldsymbol{\theta}^*, \psi^*, k)$.
Augmentation diversity
Warm-up classifier $\boldsymbol{w}_0 \leftarrow (\boldsymbol{w}; \boldsymbol{D}^{syn}, aug, opt)$.
(Phase II) Private training
for $t = 0, 1, \ldots, T-1$ **do**
  if *Extended optimization* **then**
    Run extended methods f_g (e.g., Eq. 3)
      $\boldsymbol{w}_{t+1} \leftarrow (\boldsymbol{w}_t; \boldsymbol{D}^{pr}, \boldsymbol{D}^{syn})$.
  else DP-SGD (Eq. 2) $\boldsymbol{w}_{t+1} \leftarrow (\boldsymbol{w}_t; \boldsymbol{D}^{pr})$;
end

B.1. Classification

Private training For private learning, we adopt all the techniques of [4] with WRN-16-4. We employ the techniques such as *augmentation multiplicity* to minimize the averaged loss of various augmentations $\mathcal{L}_i(\boldsymbol{w}) = \mathbb{E}_k[\ell(\boldsymbol{w}, \texttt{aug}^k(\boldsymbol{x}_i))]$, weight standardization [16], and Exponential Moving Average (EMA). We re-implement the JAX official code of [4] in https://github.com/google-deepmind/jax_privacy and extended methods using Pytorch Opacus [22] libraries.

We present the experimental details for CIFAR-10 in Table 12 and CIFAR-100 in Table 13 with their search spaces and best hyperparameter values. All experiments are conducted with DP-SGD with momentum 0 unless otherwise specified. As private learning is hugely affected by the hyperparameter settings, we use a different search space for cold and warm settings. We calculated the noise level σ for training with the hyperparameters in Tables 12 and 13 using Opacus libraries.

Warm-up training We present the experimental details of the warm-up phase with SGD (not DP-SGD) in Table 14. We use SGD with momentum 0.9 as a default setting for warm-up training. For the warm setting, we train epochs until convergence since the number of training data of 2K is less than the warmSyn of 50K samples.

Pre-trained model We use pre-trained Vision Transformers such as DeiT, and CrossViT, with the ghost clipping methods proposed in [3] and their GitHub code from https://github.com/woodyx218/private_vision. For the cold setting, we trained the models with 5 epochs with Adam optimizer with a learning rate of 0.002. We used a batch size of 1K following the default settings in GitHub. We tested various ranges of model sizes. For the warm settings, we took a grid search on the learning rate of $\{0.0005, 0.001, 0.002\}$ on both the warm-up phase and the private training phase.

B.2. Diffusion synthesis

EDM settings We implemented EDM [12] from their official GitHub code from https://github.com/NVlabs/edm. We trained the EDM model using the base settings as reported on the official GitHub repository. For the CIFAR-10 dataset, we utilized a batch size of 512 images, distributed among four NVIDIA GeForce RTX 3090 GPUs, while maintaining the other setting as the default setting. Specifically, we use a learning rate of 10^{-3}, an EMA coefficient of 0.5, duration of 200. The detailed settings are reported in Table 7 of [12]. We sampled images with $\sigma_{min} = 0.002, \sigma_{max} = 80, \rho = 7, S_{churn} = 0, S_{min} = 0, S_{max} = \infty, S_{noise} = 1$, and a step size of 18, as the default setting. For the CIFAR-100 dataset, we employed a batch size of 1024 images, distributed among four NVIDIA A100 GPUs, while maintaining the same settings as those used for the CIFAR-10 dataset. The training took about 3 days with four NVIDIA GeForce RTX 3090 GPUs. Sampling with a step size of 18 and a batch size of 500, took less than 30 seconds per batch when using a single NVIDIA GeForce RTX 3090 GPU. Therefore, it took about one hour to sample 50K images.

In summary, we require about 10-15 times for training diffusion and 0.2-0.4 times for sampling 50K images compared to training our classifiers of [4].

DG settings We implemented DG [13] from their official GitHub code from https://github.com/alsdudrla10/DG. We trained both the classifier and discriminator for DG from scratch with synthetic data from the 2K public data. We avoided using pre-trained models from the CIFAR or Imagenet datasets to solely investigate the effects by using in-distribution data.

C. Additional Notes

C.1. DP-SGD

The noise level σ of DP-SGD is determined by the total steps, sampling probability, and privacy budget (ε, δ) as follows:

Table 12. Hyperparameters for CIFAR-10.

Setup	Hyper-parameter	Search space	Best values				
Warm	ε	$\{1, 2, 3, 4, 6\}$	1	2	3	4	6
	δ	$\{10^{-5}\}$	10^{-5}	10^{-5}	10^{-5}	10^{-5}	10^{-5}
	Multiplicity K	$\{16\}$	16	16	16	16	16
	Batch size	$\{4096\}$	4096	4096	4096	4096	4096
	Clipping norm C	$\{1\}$	1	1	1	1	1
	Epochs	$\{15, 20, 30, 40\}$	15	30	30	30	20
	Learning rate η	$\{0.1, 0.5, 1, 2, 4\}$	0.5	0.5	1	1	2
+ Extended	Public batch size	$\{32, 64, 128\}$	64	64	64	64	64

Table 13. Hyperparameters for CIFAR-100.

Setup	Hyper-parameter	Search space	Best values			
Warm	ε	$\{1, 2, 6, 10\}$	1	2	6	10
	δ	$\{10^{-5}\}$	10^{-5}	10^{-5}	10^{-5}	10^{-5}
	Multiplicity K	$\{16\}$	16	16	16	16
	Batch size	$\{4096\}$	4096	4096	4096	4096
	Clipping norm C	$\{1\}$	1	1	1	1
	Epochs	$\{25, 50, 75\}$	25	50	75	75
	Learning rate η	$\{0.1, 0.5, 1, 2, 4\}$	0.5	0.5	1	1
+ Extended	Public batch size	$\{32, 64, 128\}$	64	64	64	64

Table 14. Hyperparameters for warm-up phase on CIFAR-10 and CIFAR-100.

Hyper-parameter	Search space	CIFAR-10	CIFAR-100
Batch size	$\{64, 100, 128, 256\}$	64	64
Epochs	$\{50, 100, 200\}$	100	200
Learning rate η	$\{0.05, 0.1, 0.2, 0.3\}$	0.1	0.1
Momentum	$\{0.9\}$	0.9	0.9
Learning rate decay	{Cosine}	Cosine	Cosine
Weight decay	$\{5 \times 10^{-4}\}$	5×10^{-4}	5×10^{-4}
Radius ρ for SAM	$\{0.05, 0.1, 0.2\}$	0.1	0.1

Proposition C.1 *(Abadi et al. [1]). There exist constant c_1 and c_2 so that given total steps T and sampling probability q, for any $\varepsilon < c_1 q^2 T$, DP-SGD guarantees (ε, δ)-DP, for any $\delta > 0$ if we choose*

$$\sigma \geq c_2 \frac{q\sqrt{T \log(1/\delta)}}{\varepsilon}. \tag{4}$$

C.2. Deep Analysis for Toy Example in Section 3.2

For the spiral dataset, the radius increases proportionally with the angle and at each location, the points have a probability distribution that decreases proportionally with the cumulative sum. In other words, points closer to the origin have higher probabilities. With this dataset, we construct a simple diffusion model with time step 20 and two diffusion blocks, containing a linear layer with 64 units. We train the model 10K epochs with a learning rate of 0.001 using Adam optimizer. For classifiers, we train 2-layer neural networks with ReLU, optimized using BCELoss and a learning rate of 0.1 with SGD until convergence.

With the same examples of the spiral dataset, we plot the probability density functions along the x-axis in Figure 6, with 100% (sufficient), and 10% and 4% (insufficient) cases. Specifically, the diffusion model effectively approximated the distribution, regardless of the number of training data in region A. However, in region C, where the true density is high, the generated distribution shows a higher prob-

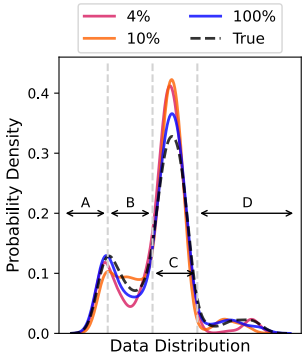


Figure 6. Probability density of toy experiment.

ability when the number of training data is limited. This indicates that when the number of training data is small, the model tends to memorize the training dataset. Conversely, in regions B and D, where the true density is low, the probability distribution of generated samples is lower than the true distribution, even if the true density is low. This indicates that when the number of samples is small, the model tends to ignore the tail distribution. As the number of in-distribution data is small, we should be aware of using synthetic data in terms of data memorization and ignore the tail part.

GAN mode collapse The diversity problem is more significant in other generative models, such as generative adversarial networks (GAN) [7]. As GAN can generative images with high fidelity, however, GAN suffers from training instability to control both generator and discriminatory, sometimes generating only a small portion of data repeatedly called mode collapse. The mode collapse happens when the convergence speed of the discrimination is faster than that of the generator, which induces the generator to generate the same images which can confuse the discriminator without generating similar images to real data. Figure 7 illustrates that the mode collapse happens with a smaller number of samples on Ring data. Thus, we mainly focus on diffusion models in this paper.

C.3. Notes on Discriminator Guidance [13]

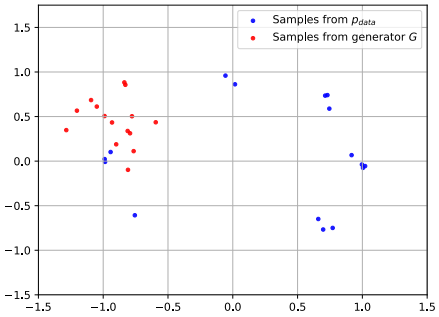
We use Discriminator Guidance (DG) [13] as a tool to understand the important factor of diffusion for performance gain without privacy concerns. By adjusting the weight of DG w_d, we determined that diversity is the key factor for sparse data. Previously, to control the trade-off between fidelity and diversity in diffusion models, Dhariwal and Nichol [5] suggested using the classifier to diffusion networks. The classifier, which is trained on noisy images during diffusion steps and their labels, can force the model to generate certain types of images based on their labels. To push further, DG [13] adopted another network called discriminator. The discriminator is trained to decide whether the images during the diffusion process are generated from real data or not. Thus, similar to the discriminator in GAN, the model can force the diffusion model to generate more similar images to the real datasets. Both studies enable the users to control the level of fidelity and diversity in diffusion sampling, where the optimal FID is obtained with a moderate level of fidelity and diversity. Instead, we focus on a high level of weight in the discriminator to generate images with high diversity, rather than repeating typical images with high fidelity.

Generation diversity and augmentation diversity We demonstrate the additional experiments on both using generation diversity by DG and augmentation diversity. Figure 8 shows CAS and private classification accuries (ε=2) for $w_d = [0, 3, 10, 20]$ for DG with or without traditional augmentation on CIFAR-10. The combination of DG(w_d=10)+Aug performs best (85.93%), surpassing EDM+Aug (85.48% in Table 6). Yet, we conclude that the gain is marginal than DG alone (2.53%p). Due to DG's extra complexity, we rather use EDM alone for experiments while using DG for investigating the important factors in generation. We believe that EDM can cover the data diversity (more than DDPM) in ID public synthesis. Note that data diversity remains a critical factor in enhancing performance when the model is not trained on specific in-distribution public data, as demonstrated in Table 10.

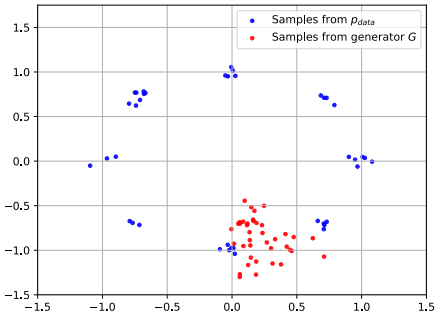
C.4. Training Dynamics

For a detailed analysis of the learning dynamics of Section 5.3, we show additional results of the private training phase. We measure the gradient norm and loss of private and synthetic data, as shown in Figure 9. As DP-SGD focuses on private data, the loss of private data decreases constantly, while the loss of public data is increasing. On the other hand, DOPE-SGD minimizes the loss of both synthetic and private data. However, in a certain range of training, DOPE-SGD occasionally fails to converge and thus diverge, where their gradient norms explode and the accuracy plummets to 0. As private training usually uses a larger learning rate than standard training, explicitly minimizing the public gradient might be dangerous as the model can be overfitted to training data and stuck into sharp minima during training.

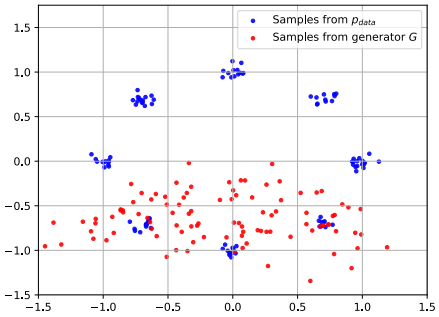
Interestingly, Figure 9d illustrates the private gradient norm without clipping. Both DOPE-SGD and DP-SGD show diverging gradient norms during training. This indicates that private training does not decrease the gradient norm, but rather decreases the clipped gradient norm.



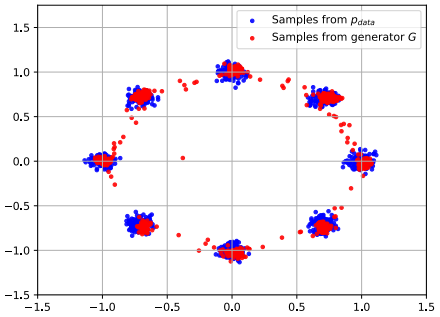
(a) GAN trained with 160 samples.



(b) GAN trained with 400 samples.



(c) GAN trained with 1K samples.



(d) GAN trained with 10K samples.

Figure 7. Mode collapse of GAN with a different number of samples on Ring data.

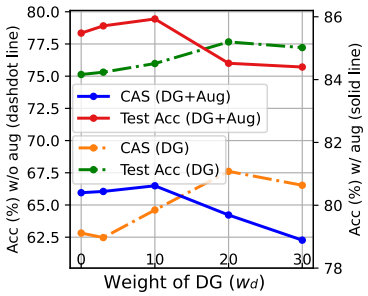


Figure 8. CAS and test accuracies by different synthetic images on the guidance weight w_d.

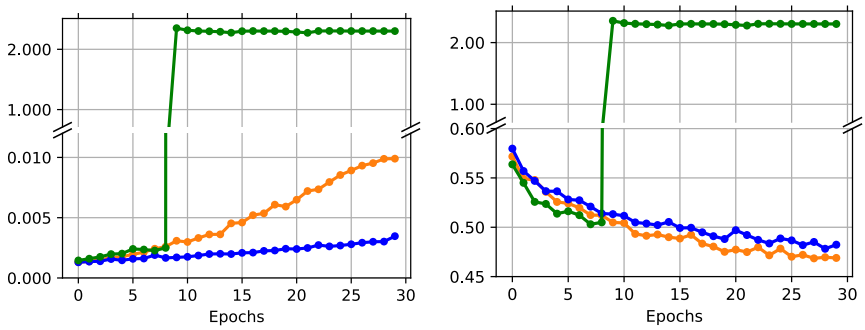
D. Ablation study

D.1. Synthetic Public Data using Text-to-Image Diffusion Models

To mitigate the drawbacks of the proposed method concerning training diffusion models, we conduct experiments with the pre-trained Stable Diffusion (SD) [6, 17], without training diffusion models. The text-to-image diffusion model is trained on a larger public LAION dataset [18]. Bao et al. [2] already utilized that using pre-trained diffusion models does not incur privacy leakage for differentially private classification tasks.

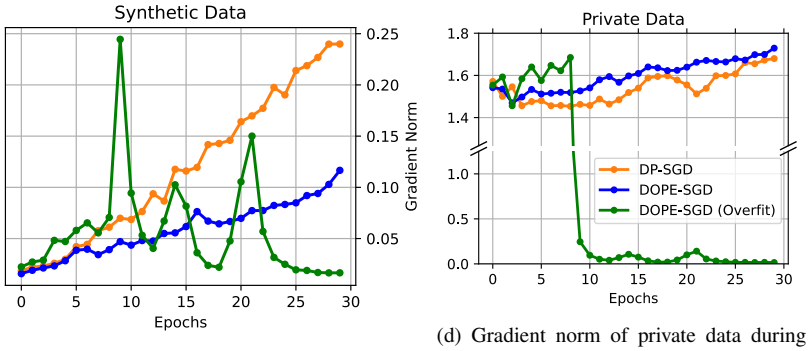
In Table 10, we examine various datasets including EuroSAT [10], which focuses on land use and land cover classification; PathMNIST [21], representing colon pathology datasets; and [11] for the case of sensitive attributes, which includes human face images depicting seven distinct race groups: White, Black, Indian, East Asian, Southeast Asian, Middle Eastern, and Latino. Without any additional fine-tuning, we provide the text prompt 'a color photo of the face of {*group name*} race for the FairFace datasets, and 'a photo of a {*class name*}.' for the other datasets to generate synthetic datasets. The data samples are resized to the size of 32×32 for training. As a result, Table 10 illustrates the strength of diffusion and diversity for warm-up training.

While we highlight the effectiveness of synthesis on a small fraction of public data, someone might argue that using ID public data might be dangerous for privacy. Thus, we conduct experiments with the Stable Diffusion without utilizing any ID public data. We compare test accuracies after private training, whether pre-trained on synthetic data or not. The results presented in Table 15 indicate that warm-up training using pre-trained diffusion models can notably enhance classification performance. This gives strong evi-



(a) Loss of synthetic data.

(b) Loss of private data.



(c) Gradient norm of synthetic data during training. training **without clipping**.

(d) Gradient norm of private data during

Figure 9. Learning dynamics during private training of various optimization methods.

Table 15. Comparison of using synthetic data from pre-trained text-to-image diffusion models, without using in-distribution public data.

Dataset	Syn	Test Acc (%)		
		$\varepsilon = 2$	$\varepsilon = 6$	$\varepsilon = 10$
CIFAR-100	✗	18.19	33.61	39.09
	✓	**52.57**	**58.94**	**60.85**
FairFace	✗	39.09	44.39	45.34
	✓	**48.21**	**53.32**	**54.75**

dence that private training, which suffers from the lack of training data samples, can be improved using diffusion synthesis. Note that the accuracies of synthetic images after the warm-up ($\varepsilon = 0$) are 29.25% for CIFAR-100 and 28.46% for FairFace. The resulting images are presented in Figure 11 of Appendix E.

D.2. Effect of Generated Data Size

To analyze the effect of the generated data size, we train models with 5K, 20K, and 40K synthetic samples in the warm-up phase until convergence for each model. The results in Table 16 indicate that data size remains a critical factor, even when applying the identical EDM model. Note that the 40K sample size is equal in the experiments of [15].

Table 16. Performance by the different number of synthetic data on CIFAR-10.

Generated	Test Acc (%)		
	$\varepsilon = 2$	$\varepsilon = 4$	$\varepsilon = 6$
5K	77.50±0.03	77.97±0.02	78.22±0.31
20K	83.21±0.12	84.05±0.04	84.16±0.08
40K	85.11±0.05	85.66±0.07	86.04±0.01

Table 17. Performance by using different amounts of public data of CIFAR-10.

Public Size	Method	Test Acc (%)		
		$\varepsilon = 2$	$\varepsilon = 4$	$\varepsilon = 6$
500 (1%)	[15]	68.9	72.1	77.1
	Ours	**78.32**	**80.99**	**82.15**
2K (4%)	[15]	75.1	77.9	80.0
	Ours	**85.48**	**86.49**	**87.06**

D.3. Different Amount of Public Data

To analyze the effects of varying amounts of public data on synthesis, we reproduce the CIFAR-10 results using only 500 public samples (1% of the training data), with each class comprising only 50 data points. We keep all other hyperparameters the same, including diffusion synthesis,

Table 18. Performance comparison of trained on the public and augmented data on CIFAR-10.

	$\varepsilon = 0$	$\varepsilon = 2$	$\varepsilon = 4$	$\varepsilon = 6$
Warm (w/ Aug and Opt)	69.63	75.44	77.22	77.99
Ours	**80.72**	**85.48**	**86.49**	**87.06**

Table 19. Test accuracy (%) of private classification using pre-trained models.

Datasets	Privacy budget		$\varepsilon = 1$			$\varepsilon = 4$		
	Architecture	Pre-trained	Cold	Warm	Ours	Cold	Warm	Ours
CIFAR -100	CrossViT small 240 (26.3M)	✓	66.88	74.70	**78.91**	72.91	78.80	**81.75**
	CrossViT 18 240 (42.6M)	✓	71.27	78.60	**81.39**	76.22	81.73	**83.75**
	DeiT base patch16 224 (85.8M)	✓	63.84	81.55	**81.62**	72.84	**84.60**	84.43
	CrossViT base 240 (103.9M)	✓	71.08	77.30	**79.65**	76.63	80.43	**82.63**

Table 20. Test accuracy of adversarial training on CIFAR-10.

Datasets	Architecture	Public	Synthesis	Methods	$\varepsilon = 2$
CIFAR -10	WRN-16-4 (2.74M)	✓	✓	Adversarial training $(8/255)$ in warm-up	77.25
		✓	✓	Adversarial training $(2/255)$ in warm-up	81.55

warm-up training, and private learning. The results presented in Table 17 demonstrate that our method consistently surpasses the previous method even when assuming a small data size.

D.4. Results of Augmentation and Generalization without Diffusion Synthesis

The results on the augmented public data and SAM, without using EDM synthesis, are presented in Table 18. The table illustrates the importance of diffusion synthesis alone without relying on augmentation and generalization.

D.5. Different Privacy Budget ε on Pre-trained CIFAR-100

We additionally append the experimental results of CIFAR-100 with pre-trained models on ImageNet, in Table 19 with privacy budget of $\varepsilon \in \{1, 4\}$.

D.6. Additional Optimization during Warm-up

We try to use other techniques for private learning to improve classification performance. The classification results with $(2, 10^{-5})$-DP are presented in Table 20 using adversarial training [14]. Adversarial training aims to make decision boundaries smooth in terms of input space. Thus, we hypothesize that using adversarial training in the warm-up training can make the private training easier. Furthermore, the importance of generative models is well under-studied in adversarial training [8, 20]. Similar to DP training, as the task of adversarial training is harder than standard training, Wang et al. [20] observed that utilizing the synthetic data with EDM can improve the generalization performance and prevent robust overfitting without extra data samples. Therefore, we adopt to use of adversarial training on the top of synthetic data in the warm-up phase. However, adversarial training methods are designed to reduce the accuracy of PGD-10 [14], and the standard accuracy is significantly dropped. Therefore, the private classification results are also decreased.

E. Generated images

We illustrate the samples of generated images using ID public data. Figure 10a demonstrates the memorization within CIFAR-10, by random sampling synthetic images in the first row and choosing the nearest samples in the synthetic dataset. Figure 10b and Figure 10c are obtained from generated images of CIFAR-10, and Figure 10d is obtained from generated images of CIFAR-100.

Furthermore, we present samples of generated images utilizing pre-trained text-to-image diffusion models in Figure 11, which is discussed in Appendix D.



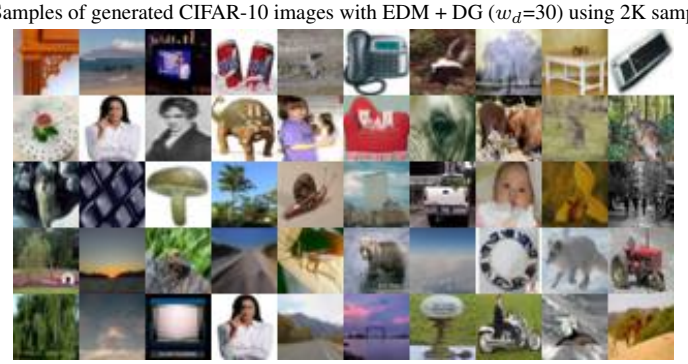
(a) Illustration of memorization within generated CIFAR-10 images.



(b) Samples of generated CIFAR-10 images with EDM using 2K samples.

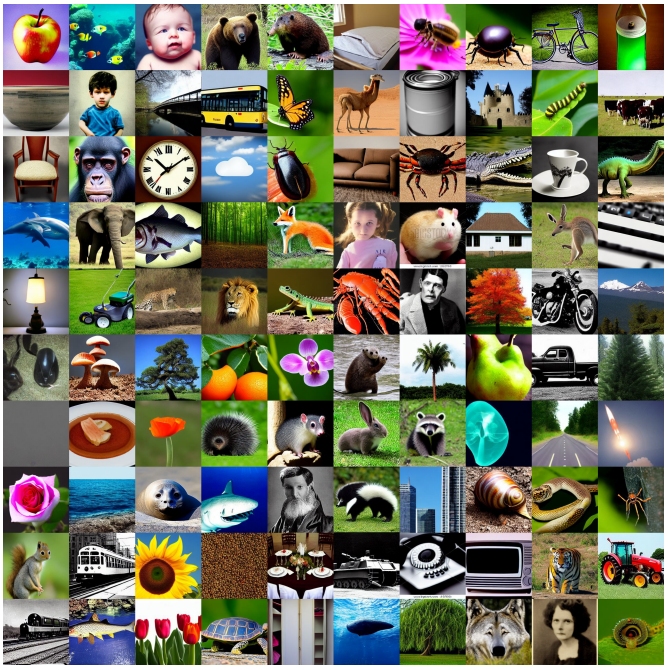


(c) Samples of generated CIFAR-10 images with EDM + DG (w_d=30) using 2K samples.

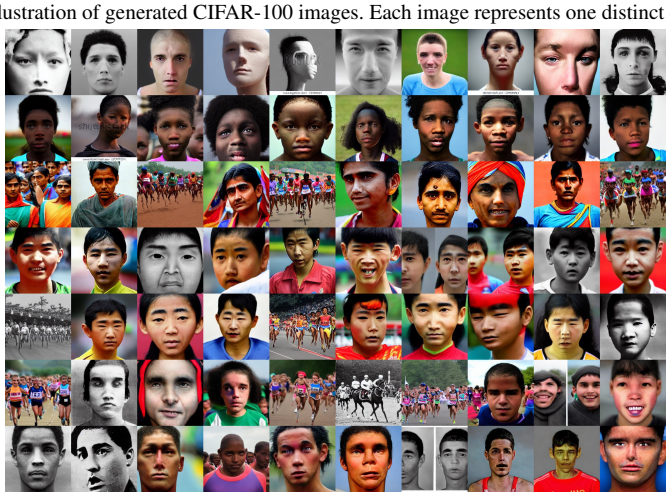


(d) Samples of generated CIFAR-100 images with EDM using 2K samples.

Figure 10. Samples of generated images using in-distribution public data.



(a) Illustration of generated CIFAR-100 images. Each image represents one distinct class.



(b) Illustration of generated FairFace images. Each row illustrates the sampled images for each race.

Figure 11. Samples of generated images using pre-trained text-to-image diffusion models.

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